

RNA regulation in host-fungal interactions: Investigation of the host epitranscriptome and fungal small RNAs

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27.02.2026

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Jena, Germany
Leibniz Institute for Natural Product
Research and Infection Biology



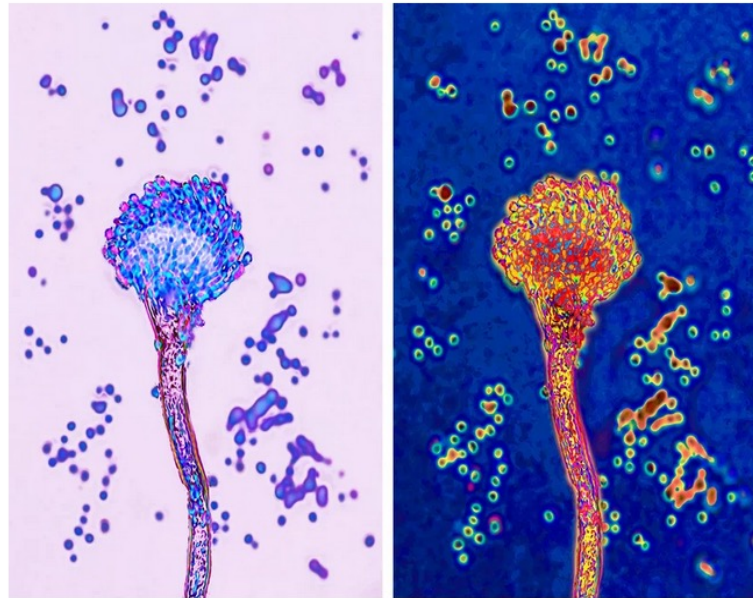
We should be worried about human fungal pathogens

WHO releases list of threatening fungi. The most dangerous might surprise you

OCTOBER 26, 2022 · 1:05 PM ET



Max Barnhart



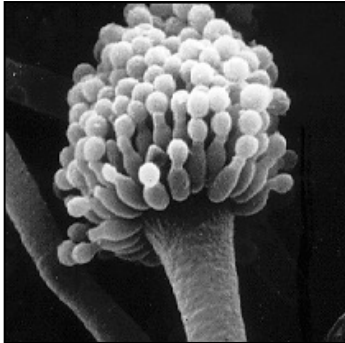
Aspergillus fumigatus can infect the lungs, causing pneumonia-like symptoms that can progress into more severe sickness.

BSIP/Education Images/Universal Images Group via Getty Images

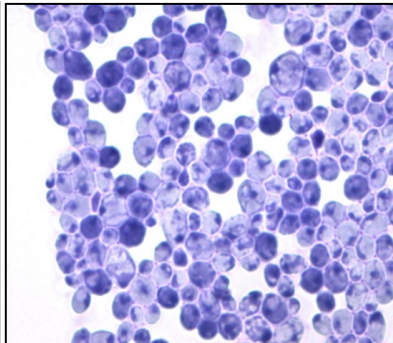
Fungi infect >1 billion people each year

And kill ~3.75 million/year

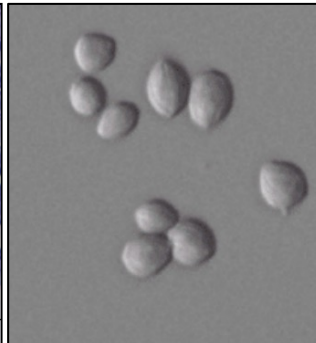
Aspergillus



Candida

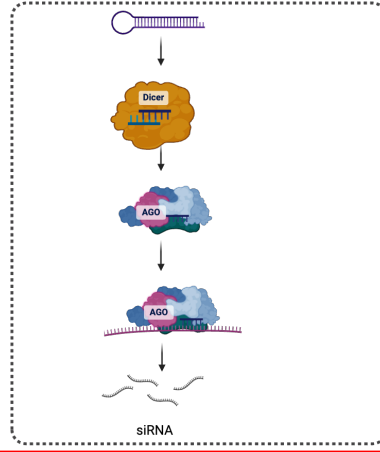


Rhizopus

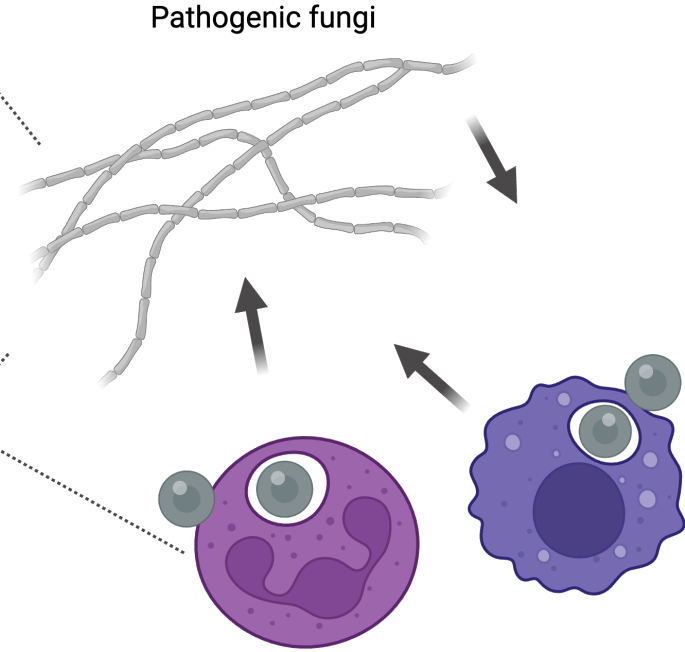
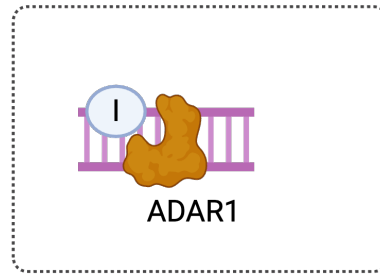


double-stranded RNA (dsRNA) metabolism during host-fungal interactions

Project A: the role of dsRNA-initiated RNA interference in pathogenic fungus *Aspergillus fumigatus*

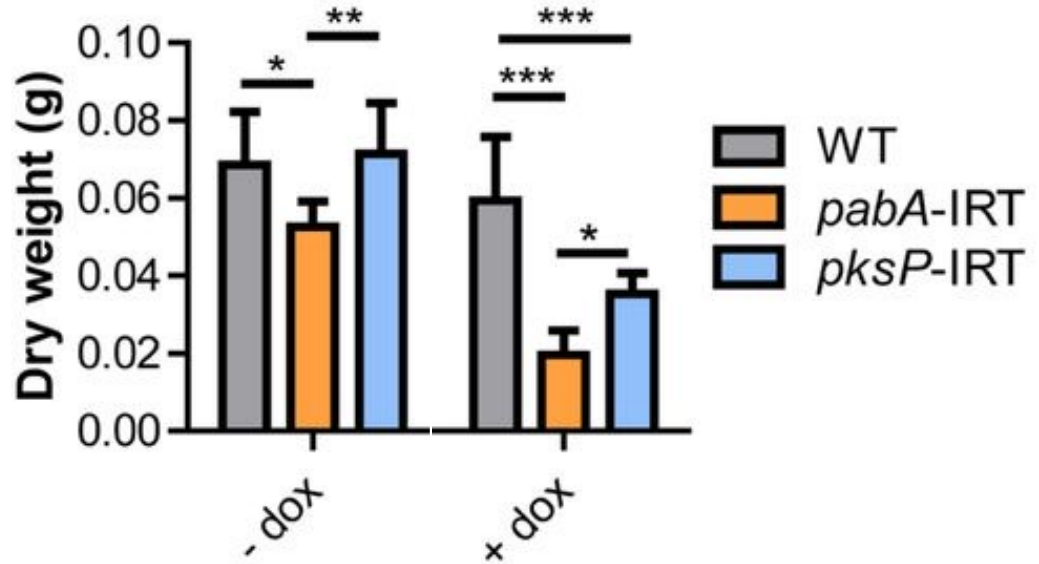
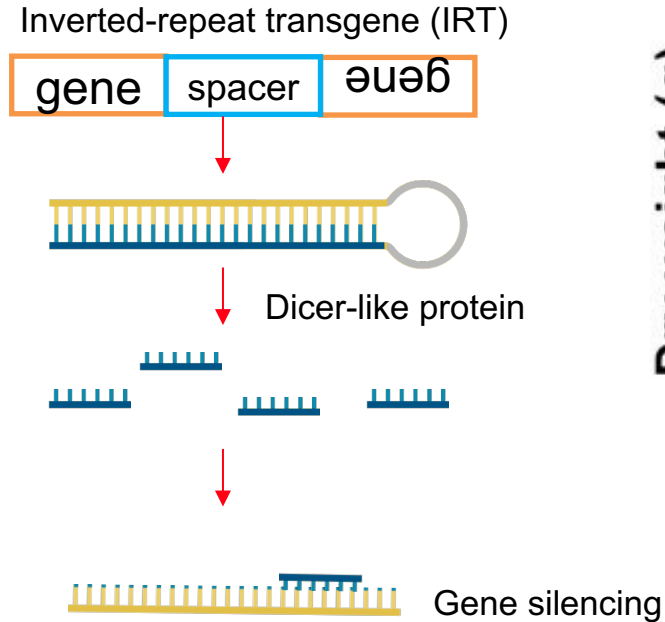


Project B: the role of dsRNA in host cells



Host cells

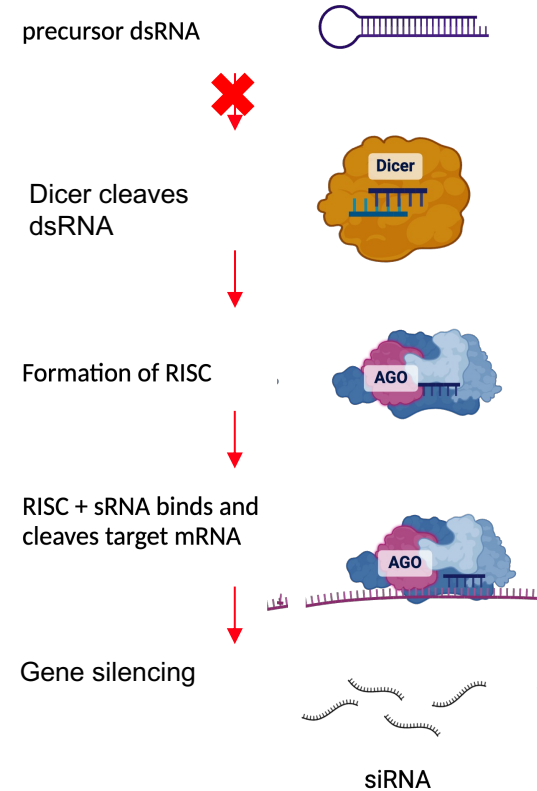
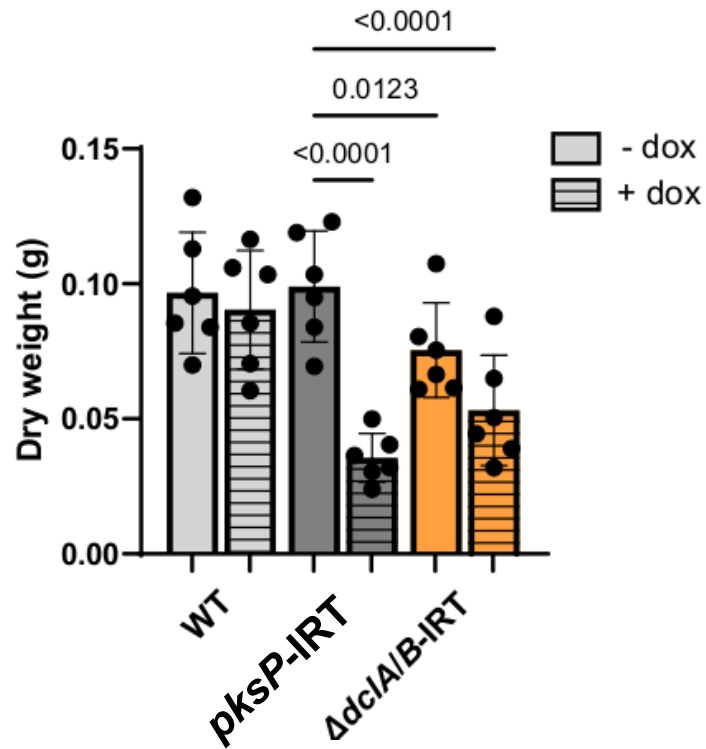
Overexpression of an IRT gene inhibit the fungal growth



conditionally essential gene (*pabA*)
nonessential gene (*pksP*)

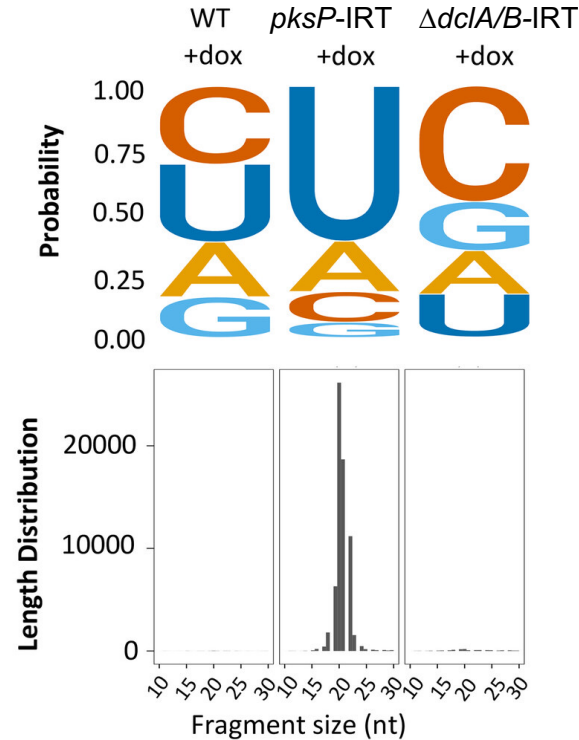
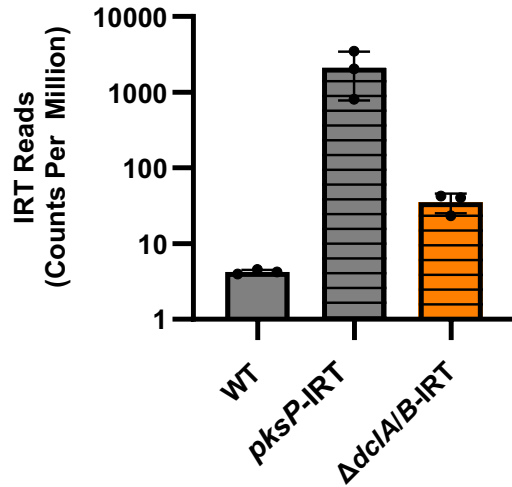
Is the fungal growth arrest
RNAi-dependent?

Fungal growth arrest is RNAi-independent



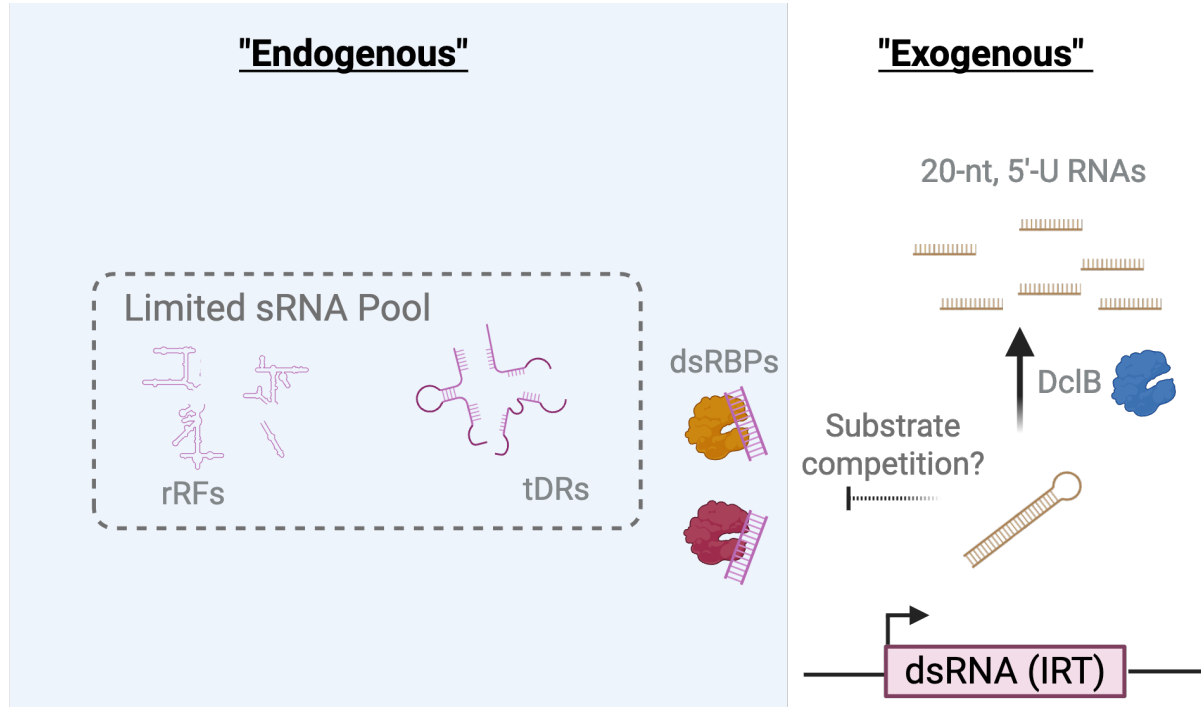
What sRNAs are produced
by RNAi in *A. fumigatus*?

IRT induces Dicer-dependent, 20-nt, 5'U sRNAs



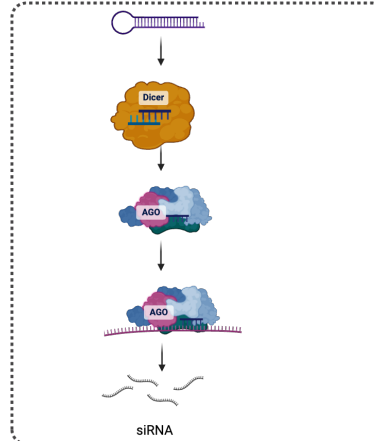
How is the IRT leading to
growth inhibition?

dsRNA overexpression limits fungal growth independent of Dicer

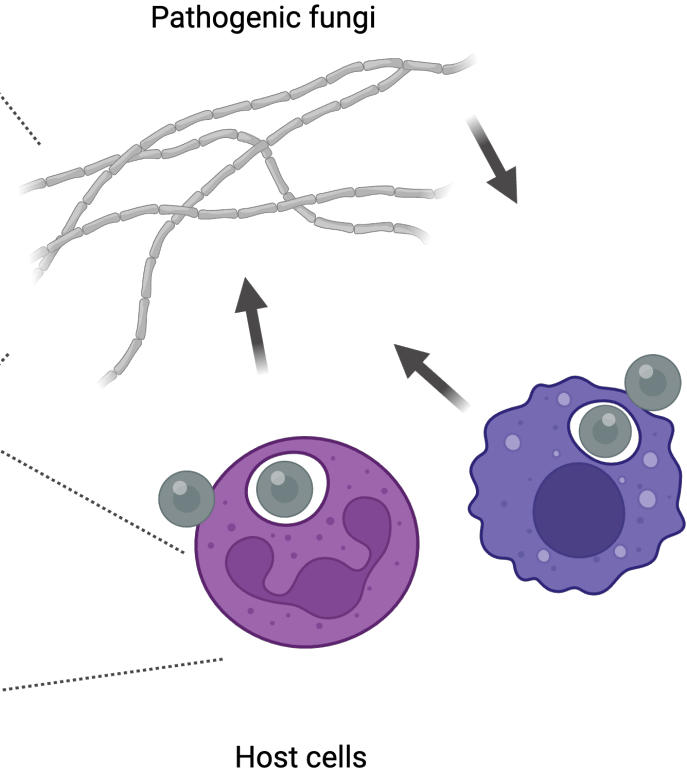
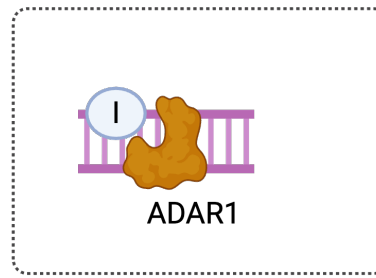


double-stranded RNA (dsRNA) metabolism during host-fungal interactions

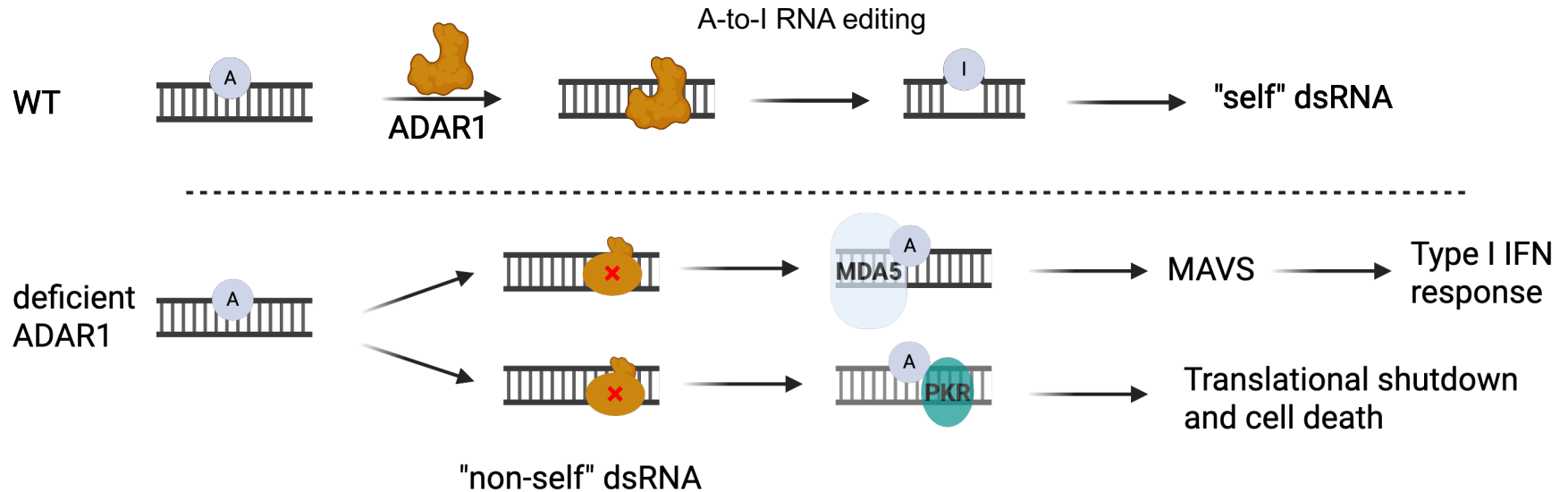
Project A: the role of dsRNA
in pathogenic fungus
Aspergillus fumigatus



Project B: the role of dsRNA
in host cells

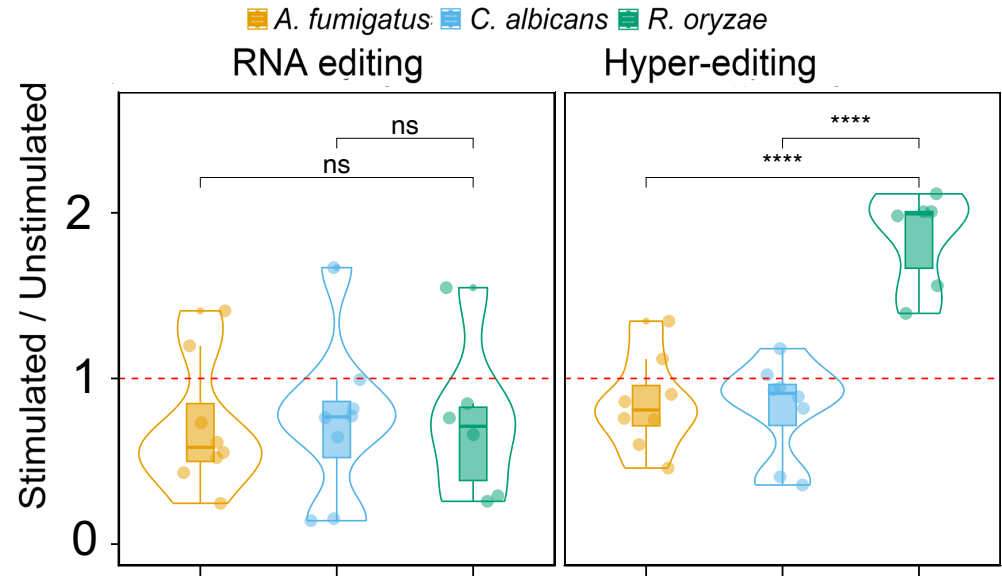
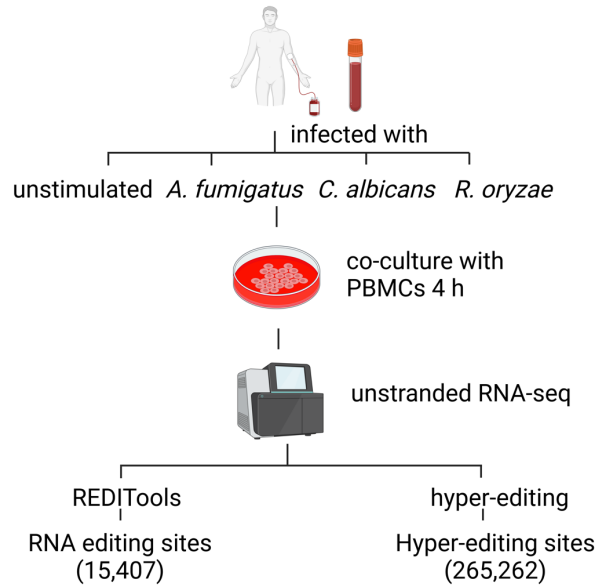


ADAR1 is an important modulator of the human immune response

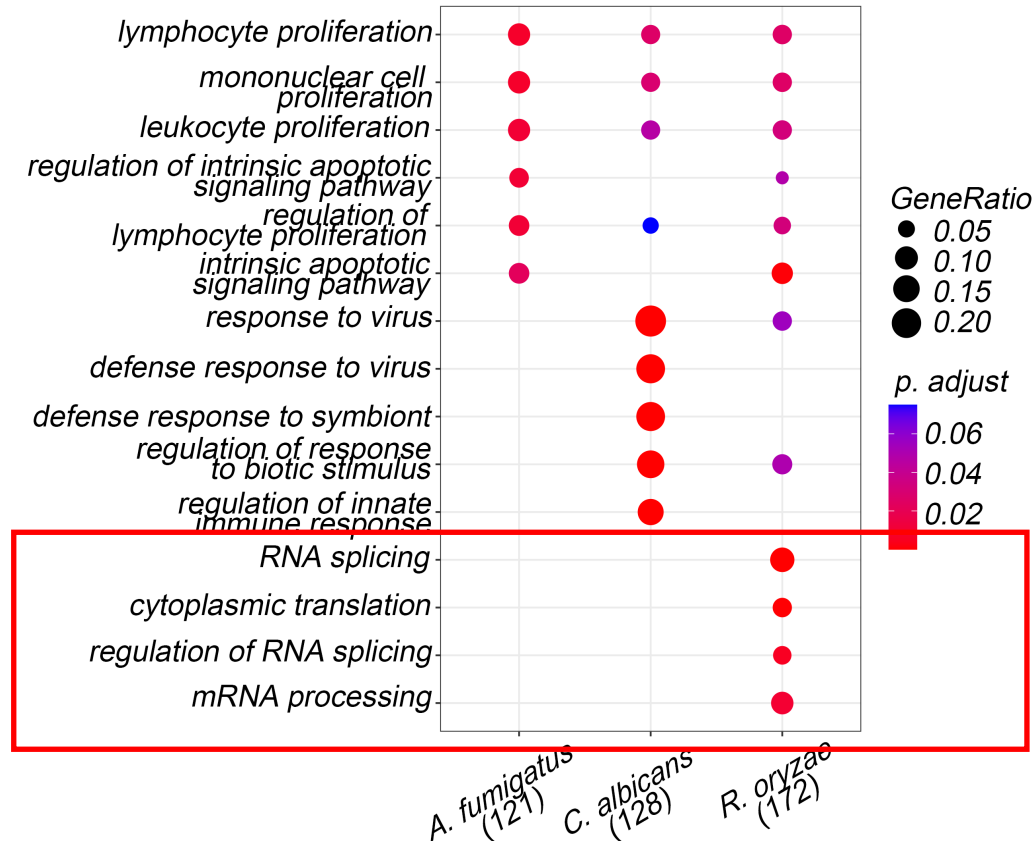


Does ADAR1 function
during fungal infections?

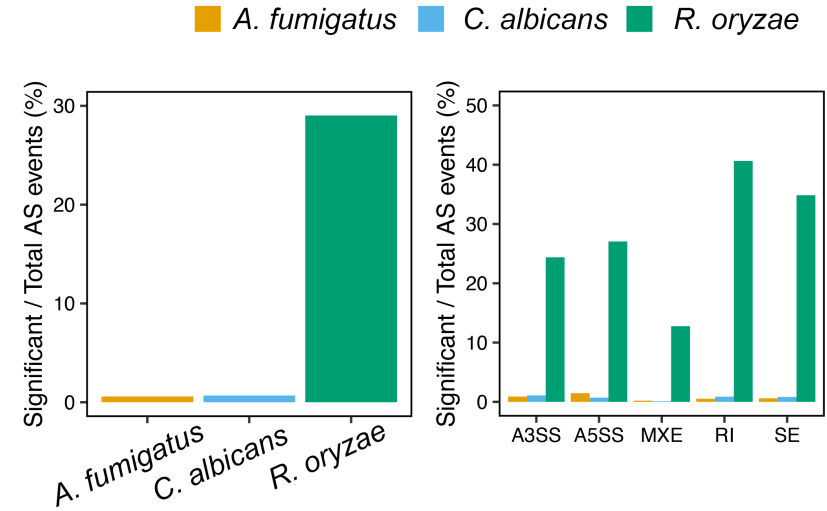
PBMCs show different RNA editing patterns upon fungal stimulations



Human A-to-I editing patterns reveal RNA splicing associated pathways upon *R. oryzae* stimulation

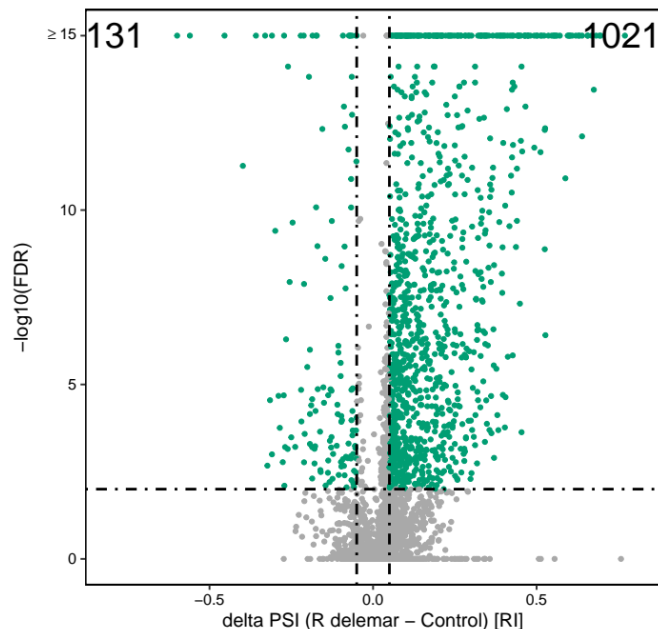
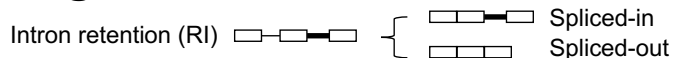


The host alternative splicing landscape changes in response to *R. oryzae* coinubation.



What is the role of
alternative splicing during
fungal incubations?

R. oryzae-induced intron retention is associated with YBX1 and SRSF2 binding motifs.



RNA binding motif analysis

Down-regulation



KHDRBS3.ATAAA[ACG]
TIA1.[AT]TTTTT[CGT]
TIA1.TTTTT[CGT][GT]
RBM47.GAT[AT]

RBM41.[AT]TAC[AT]T[GT]
HNRNPLA[AC]A[CT]A[AC]A
HNRNPL.ACAC[AG]A[ACG]
PABPC4.AAAAAA[AG]
RALY.TTTTT[CGT]

Up-regulation

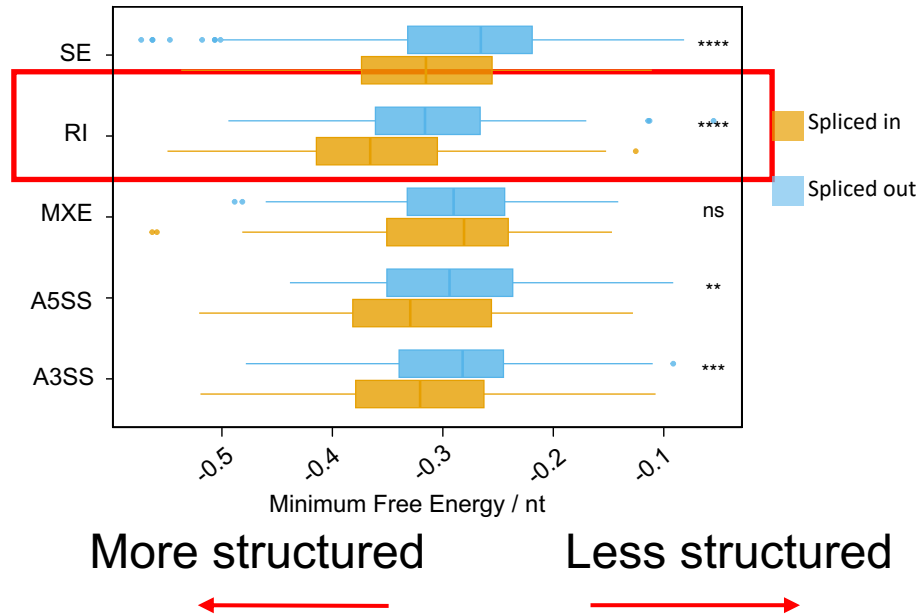


SFPQ.[GT]T[AG][AG]T[GT][GT]
YBX1.AACATC
YBX1.AACATC[AGT]
RBM3.[AG]A[AGT]AC[GT]A
SRSF2.GGAG[AT][AGT]
SRSF7.ACGACG
RBM24.[AT]G[AT]GTG[AGT]
KHDRBS2.[AG]ATAA[AC]

Increased RI



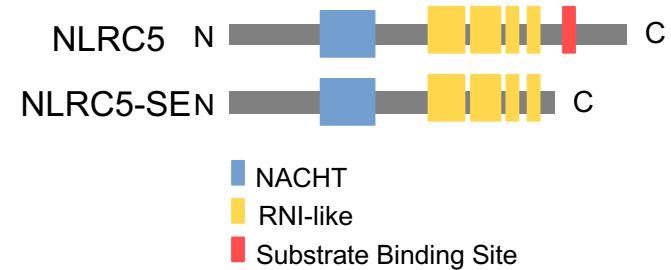
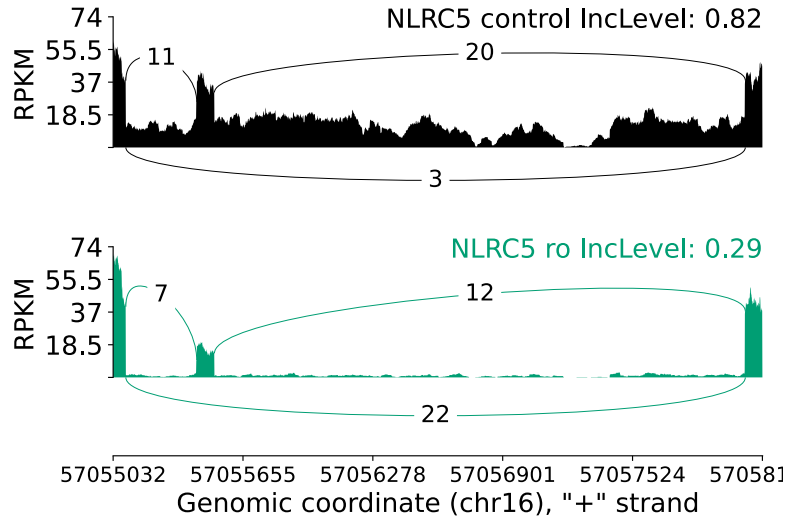
Intron retention results in more structured RNA.



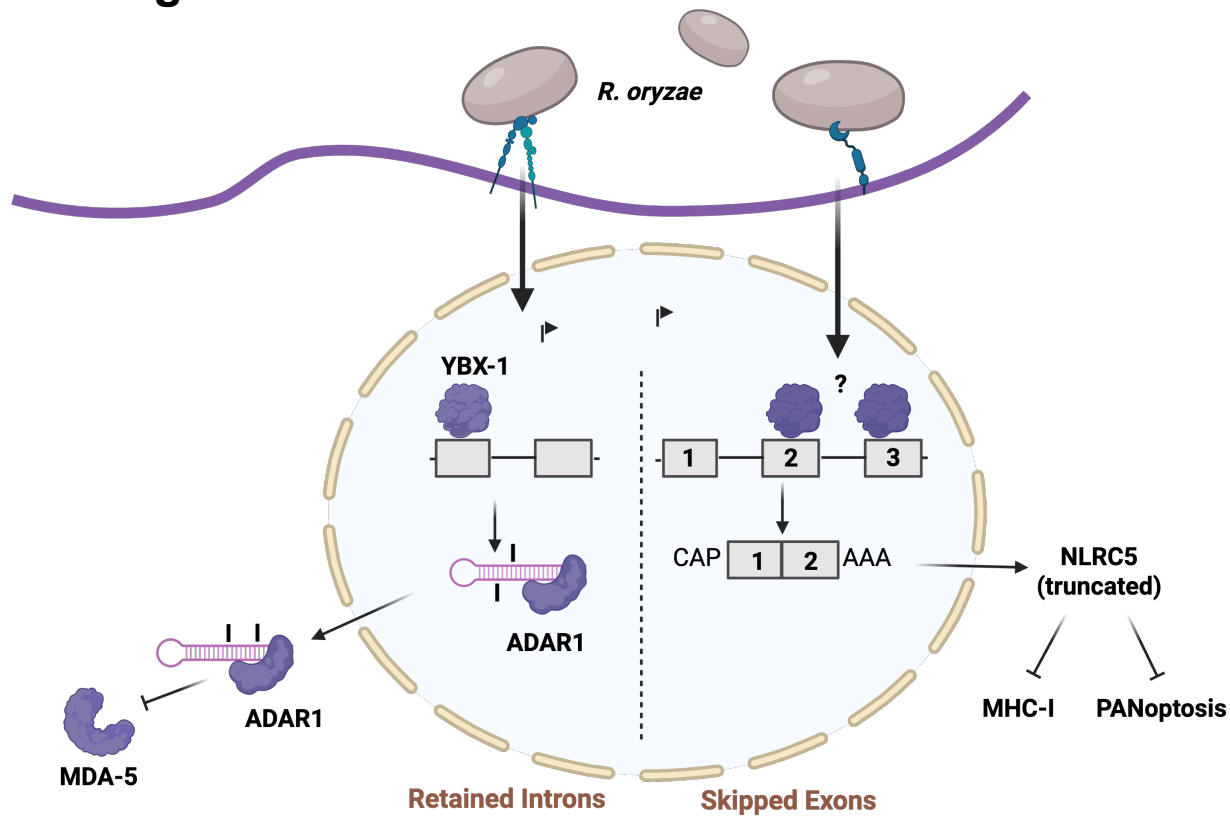
We hypothesized that structured RNA could be substrates for A-to-I editing

Exon skipping results in truncated NLRC5

NLRC5 – NOD-like receptor family CARD domain containing 5



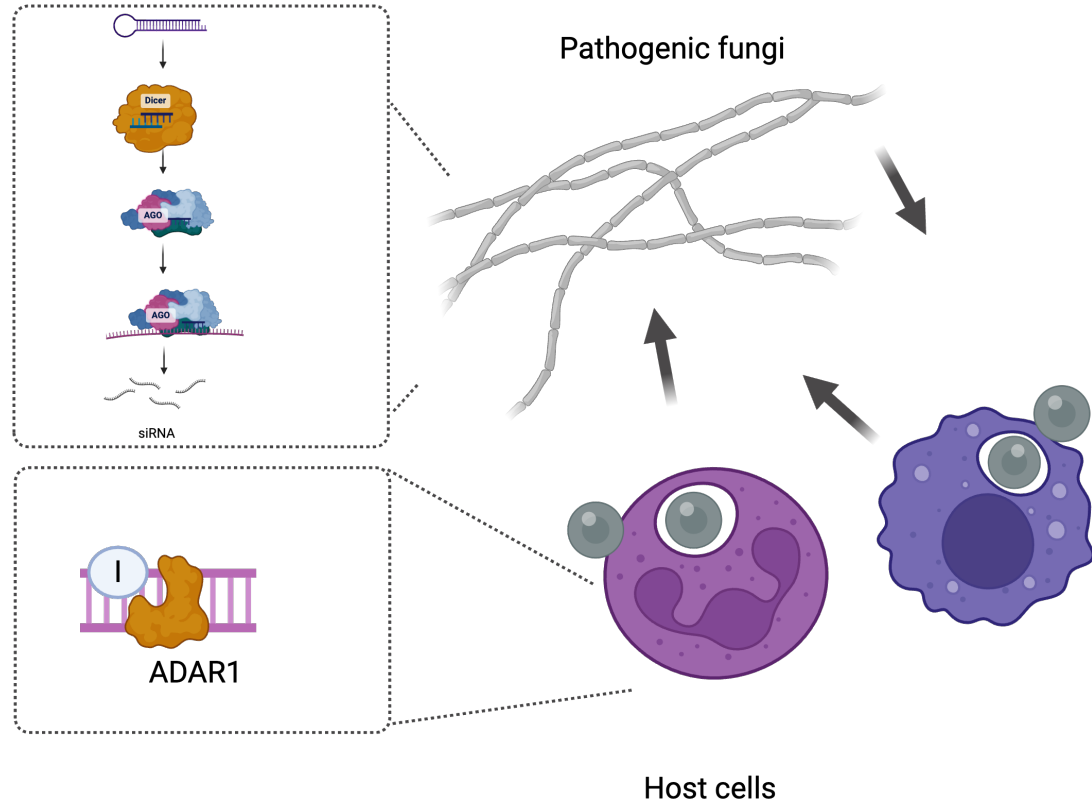
Working model



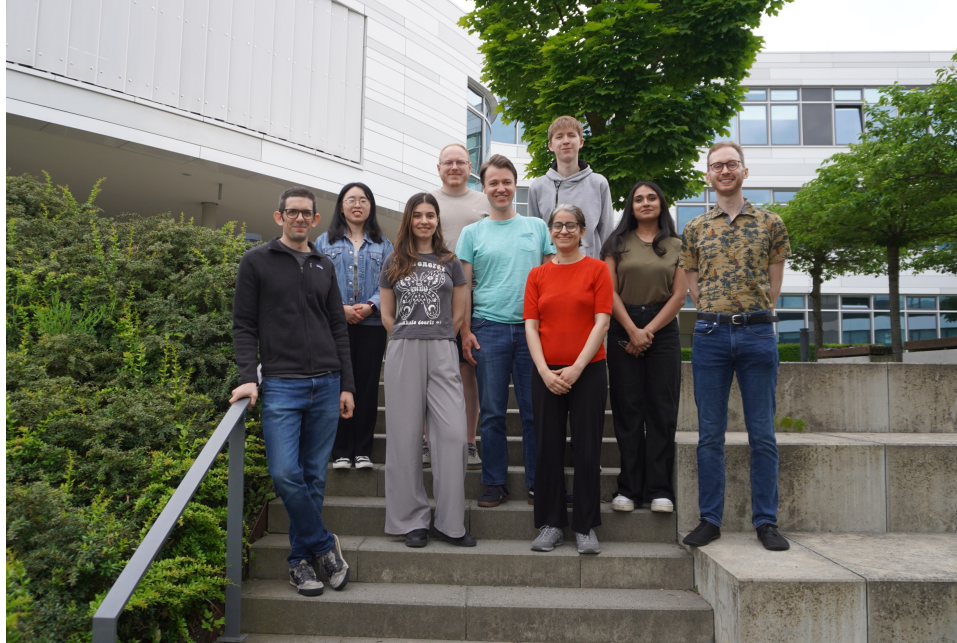
Take-home message

dsRNA overexpression results in fungal growth arrest in *A. fumigatus*.

R. oryzae incubation results in disrupted RNA splicing in PBMCs, which can further lead to ADAR1—dependent RNA editing.



Acknowledgments



RNA biology of fungal infections:

Dr. Matthew Blango

Dr. Bhawana Israni

Dr. Michael Richter

Lukas Schrettenbrunner

Valentina Lazarova

Swatika Prabakar

Akanksha Mourya

Dr. Abdulrahman Kelani

Dr. Alexander Bruch

Leibniz Institute on Aging:

Dr. Katarzyna Winek

Magnus Kummefeld

Models created with BioRender.com

GEFÖRDERT VOM



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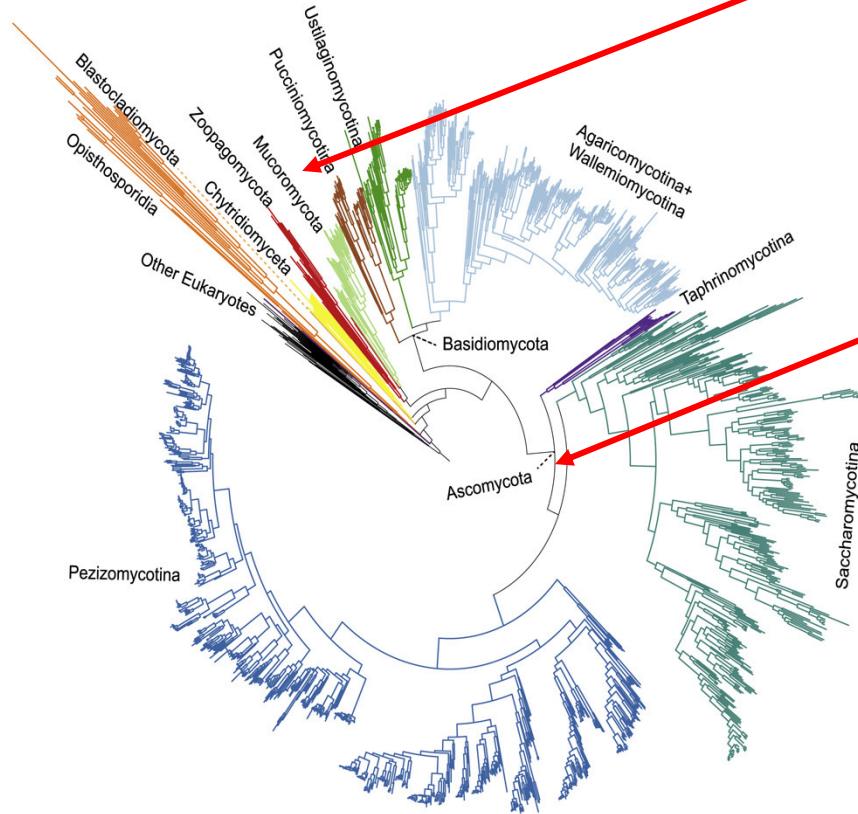
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BALANCE
OF THE
MICROVERSE

Phylogenetic tree of kingdom fungi



Rhizopus oryzae

Aspergillus fumigatus (filamentous mold)
Candida albicans (yeast)